A POINT OUTSIDE THE IMAGE OF A QUASI-COMPACT AND QUASI-SEPARATED MORPHISM IS NOT WEAKLY ASSOCIATED TO THE PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The canonical Stacks tag of the counted statement is 0AVN (divisors-lemma-weakass-pushforward). Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the tag is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. Two slips of the author inside the counted proof are reproduced verbatim: the displayed injectivity uses M_{a_r} where the preceding cover is indexed by a_1, ..., a_n, and the element m is said to lie in the stalk M_p where p should be the maximal ideal m of the local ring. No mathematical text was corrected.

Editorial note. The definition of weakly associated point is attached as a context span because the statement uses the term without repeating the definition.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of sources/stacks-chow/preamble.tex lines 14-18 and 154-217. No macro was invented or redefined.

Allowed standard prerequisites (editorial).

- Flat base change for quasi-coherent cohomology of a quasi-compact and quasi-separated morphism. Cohomology of Schemes, Lemma labelled coherent-lemma-flat-base-change-cohomology (canonical Stacks tag 02KH); invoked at proof line 736. Supplies the identification (g^* f_* F)_s = (f'_* (g')^* F)_s, which lets the proof assume that the base is local.
- The pushforward of a quasi-coherent module under a quasi-compact and quasi-separated morphism is quasi-coherent. Schemes, Lemma labelled schemes-lemma-push-forward-quasi-coherent (canonical Stacks tag 01LC); invoked at proof line 741. Lets the author write f_* F as the sheaf associated to a single A-module M, which is the object the rest of the argument estimates.
- A prime is weakly associated to a module exactly when it is the radical of the annihilator of some element. Algebra, Definition labelled algebra-definition-weakly-associated and Algebra, Lemma labelled algebra-lemma-weakly-ass-local (canonical Stacks tags 0547 and 0566), recalled in the section introducing the definition. Turns the conclusion "s is not weakly associated to f_* F" into the explicit statement that no nonzero element m of the stalk has all a_i^n m equal to zero.

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Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Cross-chapter pointer invoked at proof line 736 for flat base change; the resulting identification of stalks is then used as the reduction to a local base, which the counted proof carries out itself.

Source reference. Original lemma 01LC: sources/stacks-chow/tags.txt, source lines 2081–2081. Cross-chapter pointer invoked at proof line 741 for quasi-coherence of f_* F; the module M and every estimate on it are then produced inside the counted proof.

Lemma 0.1. *Let $f : X \to S$ be a quasi-compact and quasi-separated morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $s \in S$ be a point which is not in the image of f. Then s is not weakly associated to $f_*\mathcal{F}$.*

Proof. Consider the base change $f' : X' \to \mathrm{Spec}(\mathcal{O}_{S,s})$ of f by the morphism $g : \mathrm{Spec}(\mathcal{O}_{S,s}) \to S$ and denote $g' : X' \to X$ the other projection. Then

$$(f_*\mathcal{F})_s = (g^* f_*\mathcal{F})_s = (f'_*(g')^*\mathcal{F})_s$$

The first equality because g induces an isomorphism on local rings at s and the second by flat base change (Cohomology of Schemes, Lemma 02KH). Of course $s \in \mathrm{Spec}(\mathcal{O}_{S,s})$ is not in the image of f'. Thus we may assume S is the spectrum of a local ring $(A, \mathfrak{m})$ and s corresponds to $\mathfrak{m}$. By Schemes, Lemma 01LC the sheaf $f_*\mathcal{F}$ is quasi-coherent, say corresponding to the A-module M. As s is not in the image of f we see that $X = \bigcup_{a \in \mathfrak{m}} f^{-1}D(a)$ is an open covering. Since X is quasi-compact we can find $a_1, \ldots, a_n \in \mathfrak{m}$ such that $X = f^{-1}D(a_1) \cup \ldots \cup f^{-1}D(a_n)$. It follows that

$$M \to M_{a_1} \oplus \ldots \oplus M_{a_r}$$

is injective. Hence for any nonzero element m of the stalk $M_{\mathfrak{p}}$ there exists an i such that $a_i^n m$ is nonzero for all $n \geq 0$. Thus $\mathfrak{m}$ is not weakly associated to M. □

Definition 0.2. Let X be a scheme. Let $\mathcal{F}$ be a quasi-coherent sheaf on X.

(1) We say $x \in X$ is *weakly associated* to $\mathcal{F}$ if the maximal ideal $\mathfrak{m}_x$ is weakly associated to the $\mathcal{O}_{X,x}$-module $\mathcal{F}_x$.
(2) We denote $\mathrm{WeakAss}(\mathcal{F})$ the set of weakly associated points of $\mathcal{F}$.
(3) The *weakly associated points of X* are the weakly associated points of $\mathcal{O}_X$.